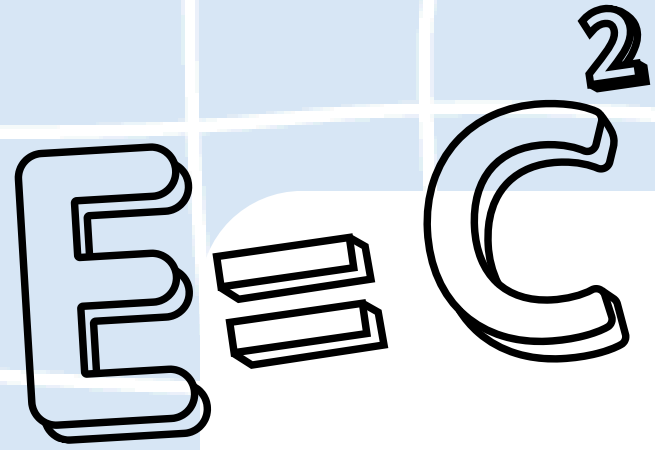
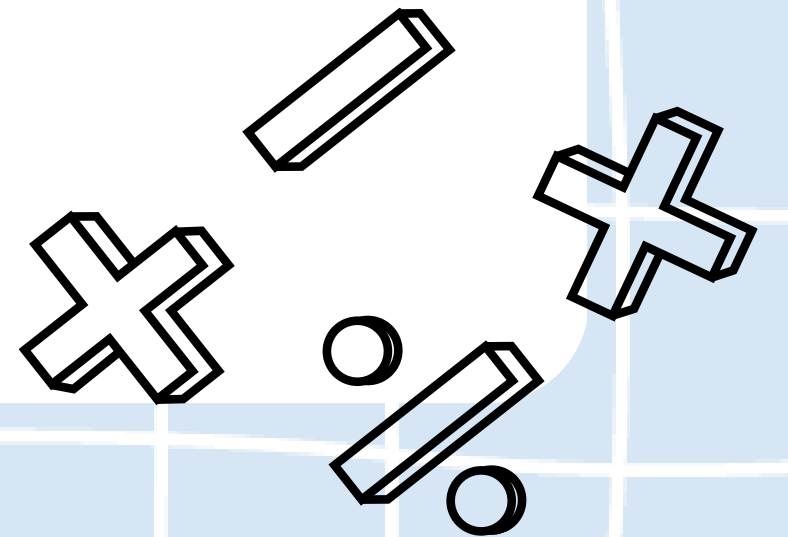




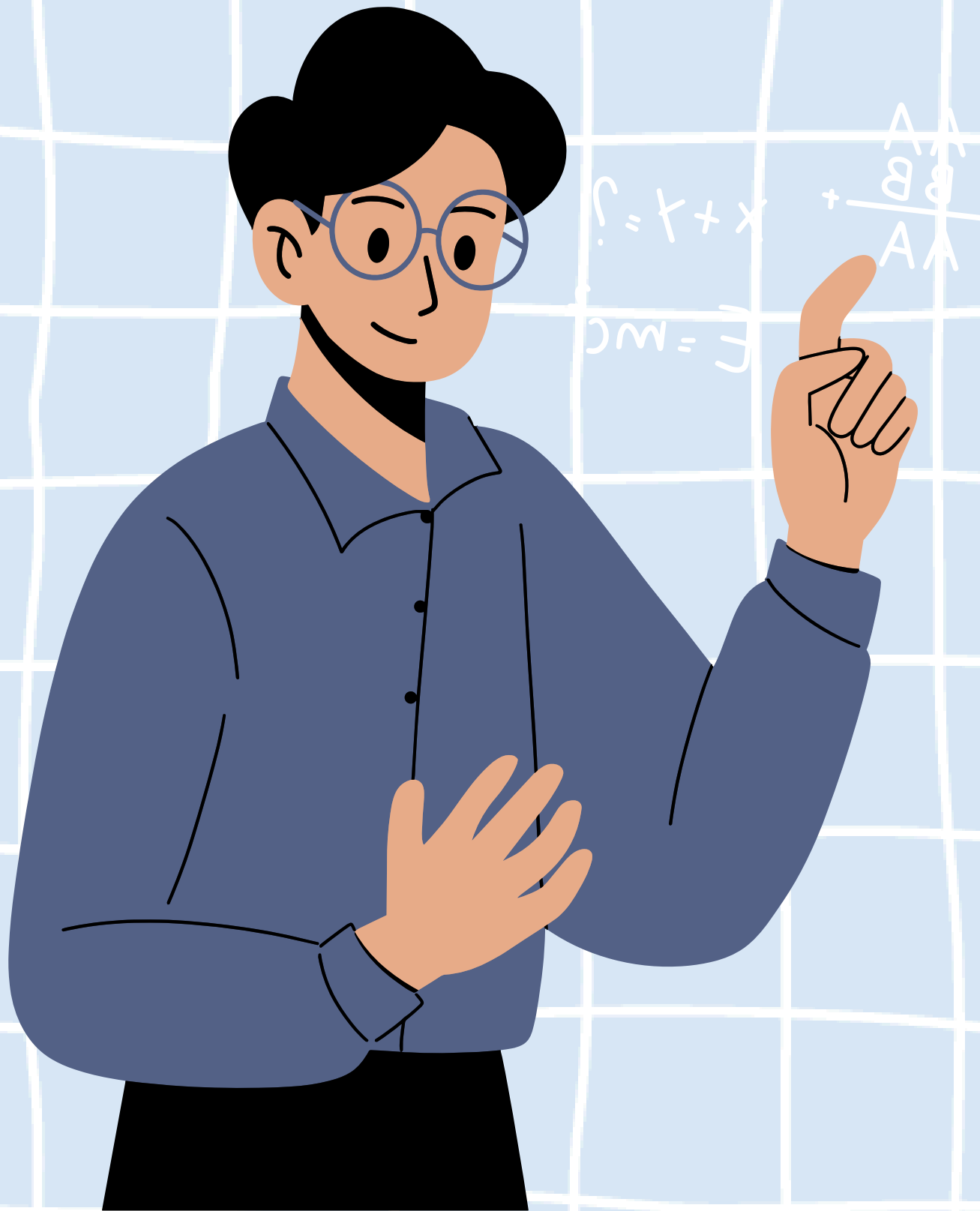
Decimals & percent

Grade 7

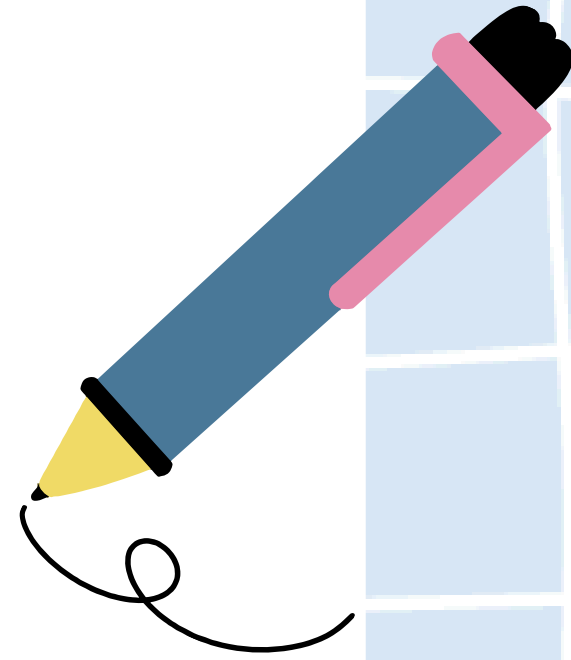


Introduction

Where have you seen
percents and decimals in
real life?

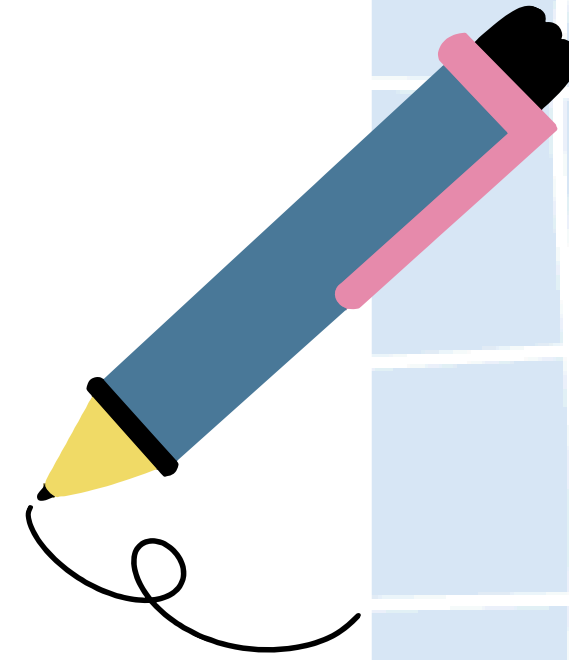


Decimals



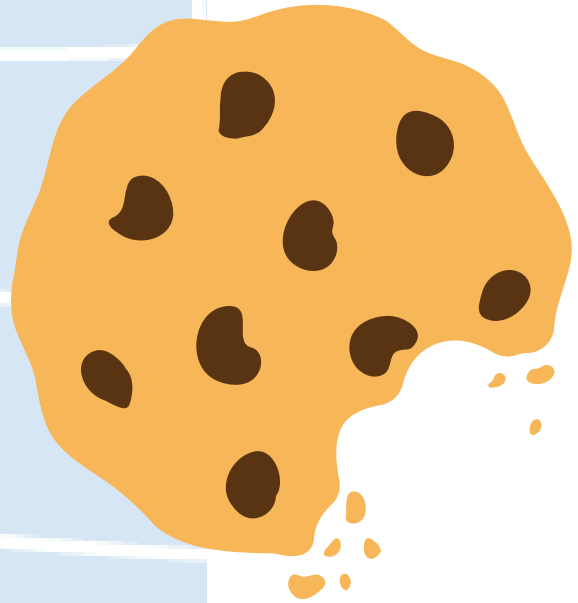
Decimals are numbers that have a **whole part** and a **part of a whole**, **separated by a decimal point**. They are used for money transactions, weight, measurements, test score or sport scores.

Percentages



Percentages are **parts per hundred**. They can be represented as fractions with a denominator of 100. Examples: discounts, tip, test scores or statistics.

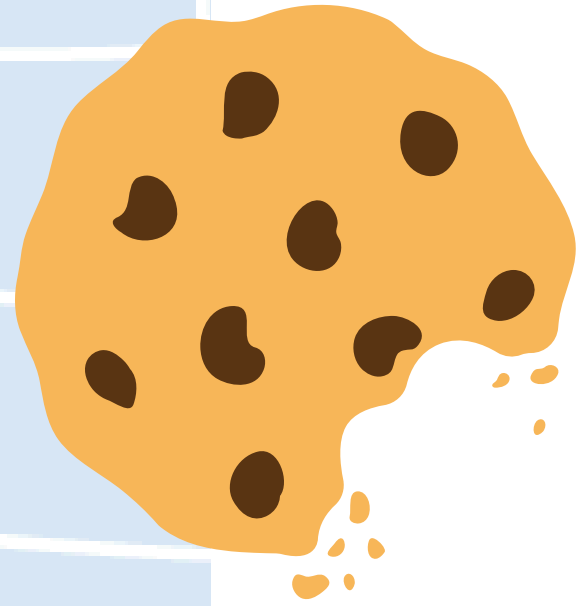
Example



If you have 10 cookies and you ate 0.3 of them, that means you ate 3 cookies because 0.3 means 3 tenths, and 3 tenths of 10 is 3.

$$0.3 = 3/10$$

$$3/10 \text{ of } 10 \text{ cookies} = 3 \text{ cookies}$$



Example

If we say 30% of the cookies
have been eaten, then
 $30/100$, or 30 cookies are
gone.

Rounding decimals

Look at the digit right after the place you're rounding to:

If that digit is 5 or more, round up

If it's less than 5, round down

Rounding to the nearest tenth:

$0.18 \rightarrow 0.2$ (8 is 5 or more)

$0.12 \rightarrow 0.1$ (2 is less than 5)

$0.15 \rightarrow 0.2$ (5 is 5 or more)

Add & subtract

1. Line up the decimal points.
2. If a number has fewer decimal places, add zeros to match.
3. Then add or subtract like normal.

$$4.236 + 3.450 = 7.686$$

$$6.00 - 0.72 = 5.28$$

$$0.17 + 0.08 = 0.25$$

$$1.89 - 1.70 = 0.19$$



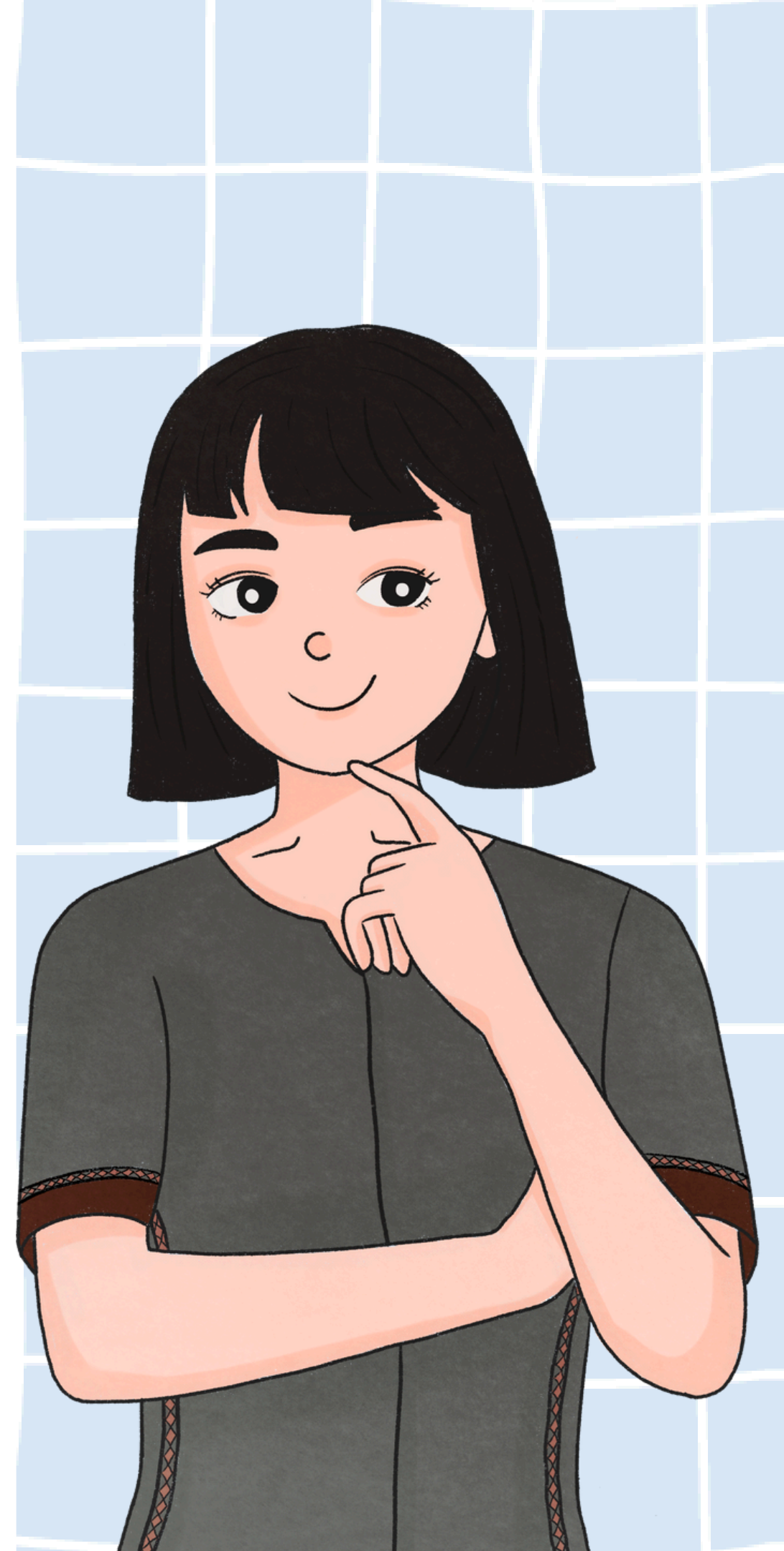
Multiply & divide

1. Count decimal places.
2. Multiply like whole numbers.
3. Move the decimal in the answer.

$$1.47 \times 0.2 \rightarrow 294 \rightarrow 0.294$$

- ÷
1. Write as a fraction.
 2. Multiply top & bottom to remove decimal
 3. Divide.

$$0.2 \div 0.8 = 2/8 = 0.25$$



Percent

A percent means “out of 100” - Example: $32\% = 32/100$

**“Percent” = specific amount & “Percentage” = general amount
(e.g., 5% of the class vs. a percentage of the class)**

Converting:

Decimal → Percent: $\times 100$

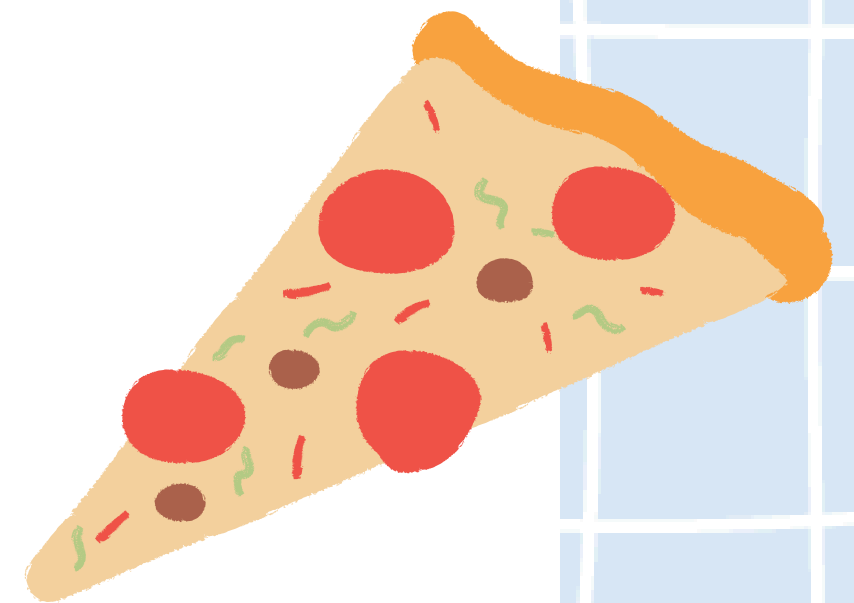
Example: $0.47 = 47\%$

Percent → Decimal: $\div 100$

Example: $125\% = 1.25$

Example

If we say 25% of a pizza has been eaten, then $25/100$, or $1/4$, of the pizza is gone.



Percent of an amount

1. Change the percent to a decimal(Divide it by 100)
2. Multiply the decimal by the total amount

✓ Example 1: What is 25% of 80?

1. $25\% = 0.25$
2. $0.25 \times 80 = 20$

Answer: 25% of 80 is 20

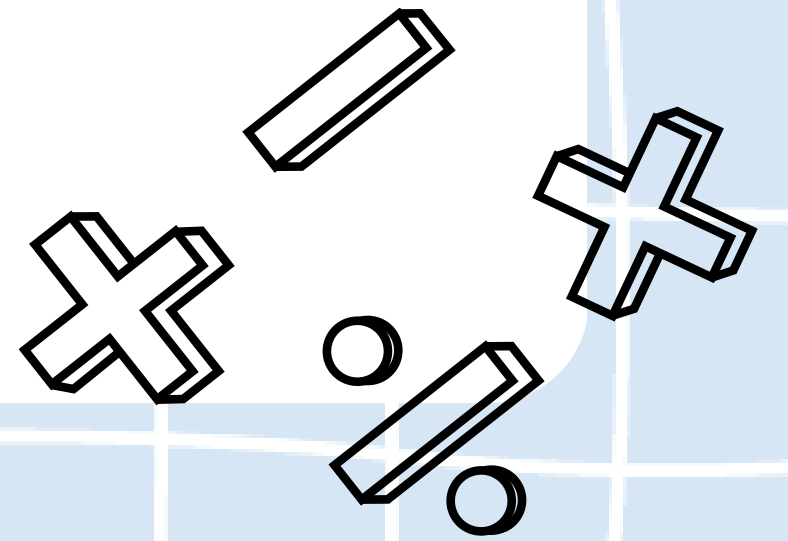
✓ Example 2: What is 60% of 150?

1. $60\% = 0.60$
2. $0.60 \times 150 = 90$



$$E = C^2$$

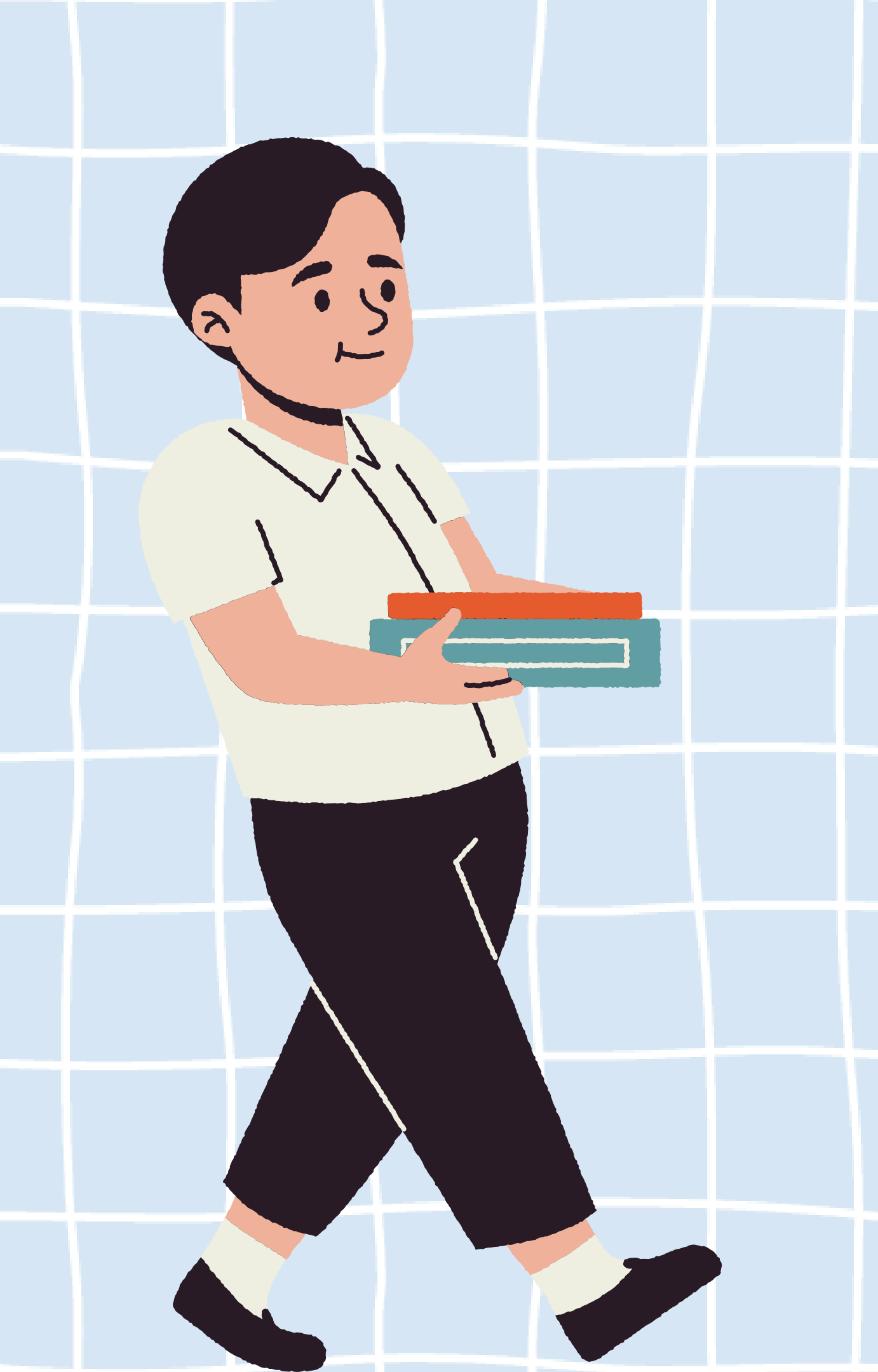
Quiz Time



Question #1

- A) Convert 2.4 to a percent
- B) What is 75% of 80
- C) A water bottle is filled to 0.65 of its total capacity.
What percent is it?





Question #2

How are percents,
fractions, and
decimals related?